

Coding & Solution

Date	29 September 2026
Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	5 Marks

Coding & Solution

The implemented solution follows the planned FastAPI + Gemini + SQLite architecture. The source package is included in the root of this submission.

Implemented flow

1. `index.html` collects user profile details. 2. `generate-workout` validates and saves the profile. 3. Gemini generates the workout and nutrition tip. 4. The plan is stored through SQLAlchemy. 5. `result.html` displays the output. 6. `submit-feedback` sends feedback to the update generator and saves the revised plan. 7. `view-all-users` displays stored user records and supports deletion.

Representative source locations

Source File	Purpose	Evidence in implementation
app/main.py	Create FastAPI app, mount static files and templates, include routes, health endpoint.	FastAPI instance and /health.
app/routes.py	Orchestrate form submissions, AI calls, storage, feedback and result rendering.	/generate-workout, /submit-feedback, /view-all-users.
app/ai/gemini_generator.py	Generate seven-day workout plan.	generate_workout_gemini().
app/ai/gemini_flash_generator.py	Generate nutrition/recovery tip.	generate_nutrition_tip_with_flash().
app/ai/updated_plan.py	Regenerate plan from user feedback.	update_workout_plan().
app/database.py	Create tables and CRUD-style helpers.	User, Plan, save_user, save_plan, update_plan.
templates/*.html	Render user form, results, admin view and errors.	Jinja2 templates.
tests/test_app.py	Automated smoke tests.	Home and health endpoint tests.

Quality / safety notes

The Gemini prompts explicitly avoid medical diagnosis, dangerous exercise, extreme dieting and unsafe rapid weight loss. The application is an educational fitness-planning prototype and not a medical assessment system.